Abstract

Abstract I

Abstract II

The reproducibility crisis in computational research is fundamentally structural: research artifacts are scattered across disconnected tools—LaTeX editors, Jupyter notebooks, ad-hoc shell scripts—with no enforced mechanism to keep code, data, and manuscript synchronized. Studies have shown that most published findings are false positives, replication rates in psychology hover around 36%, and only 24% of 1.4 million Jupyter notebooks can be successfully re-executed. Existing tools address fragments of this problem: workflow managers (Snakemake, Nextflow, CWL) orchestrate computation; literate programming systems (Quarto, Jupyter Book, R Markdown, Overleaf, OpenAI Prism) render documents; data versioning tools (DVC) track artifacts—but none enforces cross-cutting quality standards as architectural invariants. `template/` applies the principle of Infrastructure as Code to the research lifecycle, making the manuscript, test suite, and provenance chain version-controlled, deterministically buildable, and independently verifiable. It is built on a Two-Layer Architecture that separates 23 infrastructure subdirectories (20 importable Python packages, $\sim$604 modules, validated by $\sim$7,780